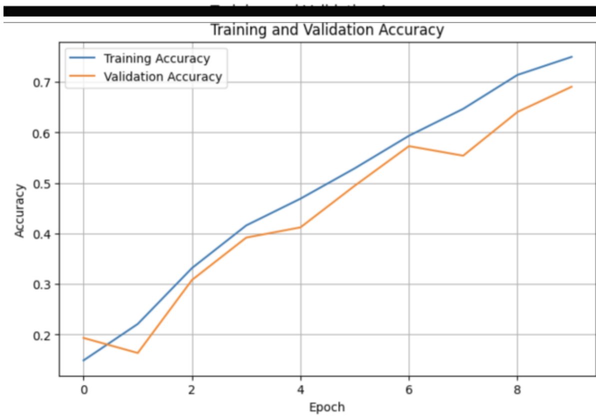
```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a layer. When
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

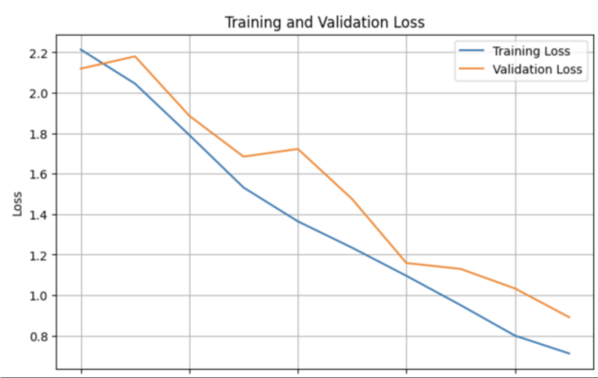
Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 30, 30, 32)	896
max_pooling2d (MaxPooling2D)	(None, 15, 15, 32)	0
conv2d_1 (Conv2D)	(None, 13, 13, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 6, 6, 64)	0
conv2d_2 (Conv2D)	(None, 4, 4, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 2, 2, 128)	0
flatten (Flatten)	(None, 512)	0
dense (Dense)	(None, 128)	65,664
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 10)	1,290

Total params: 160,202 (625.79 KB)
Trainable params: 160,202 (625.79 KB)
Non-trainable params: 0 (0.00 B)

```
================ TRAINING =================
Epoch 1/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 12s 105ms/step - accuracy: 0.1488 - loss: 2.2136 - val_accuracy: 0.1932 - val_loss: 2.1192
Epoch 2/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 9s 96ms/step - accuracy: 0.2210 - loss: 2.0451 - val_accuracy: 0.1635 - val_loss: 2.1799
Epoch 3/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 9s 81ms/step - accuracy: 0.3316 - loss: 1.7913 - val_accuracy: 0.3081 - val_loss: 1.8863
Epoch 4/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 9s 102ms/step - accuracy: 0.4160 - loss: 1.5317 - val_accuracy: 0.3919 - val_loss: 1.6843
Epoch 5/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 9s 101ms/step - accuracy: 0.4690 - loss: 1.3645 - val_accuracy: 0.4122 - val_loss: 1.7224
Epoch 6/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 8s 80ms/step - accuracy: 0.5290 - loss: 1.2343 - val_accuracy: 0.4946 - val_loss: 1.4749
Epoch 7/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 9s 99ms/step - accuracy: 0.5935 - loss: 1.0953 - val_accuracy: 0.5730 - val_loss: 1.1587
Epoch 8/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 8s 88ms/step - accuracy: 0.6468 - loss: 0.9503 - val_accuracy: 0.5541 - val_loss: 1.1293
Epoch 9/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 10s 105ms/step - accuracy: 0.7139 - loss: 0.7997 - val_accuracy: 0.6405 - val_loss: 1.0329
Epoch 10/10
93/93 ━━━━━━━━━━━━━━━━━━━━ 9s 97ms/step - accuracy: 0.7497 - loss: 0.7115 - val_accuracy: 0.6905 - val_loss: 0.8907
```

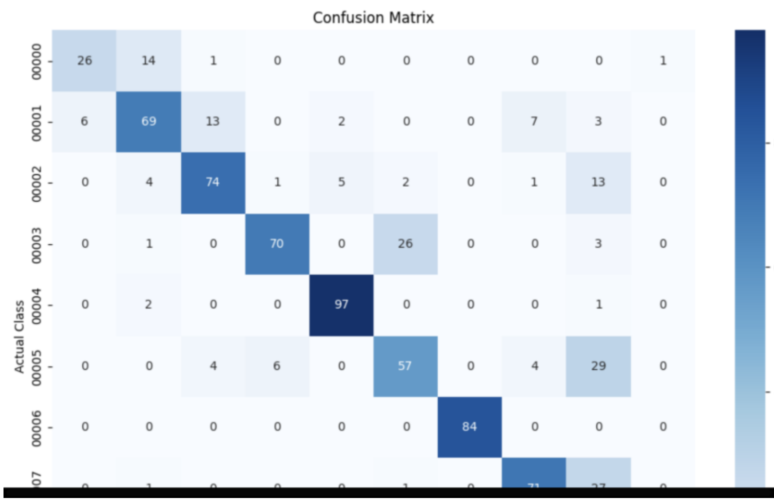




```
================ CLASSIFICATION REPORT ================
              precision    recall  f1-score   support

       00000       0.81      0.62      0.70        42
       00001       0.76      0.69      0.72       100
       00002       0.78      0.74      0.76       100
       00003       0.89      0.70      0.78       100
       00004       0.93      0.97      0.95       100
       00005       0.66      0.57      0.61       100
       00006       1.00      1.00      1.00        84
       00007       0.84      0.71      0.77       100
       00008       0.56      0.95      0.70       100
       00009       0.99      0.98      0.98       100

    accuracy                           0.80       926
   macro avg       0.82      0.79      0.80       926
weighted avg       0.82      0.80      0.80       926
```




```
=============== MODEL INFORMATION ===============
Number of Model Parameters: 160202

Model saved successfully!
Model Location: /content/traffic_sign_cnn_model.keras

====================================================
     CNN TRAFFIC SIGN RECOGNITION COMPLETED
====================================================

Final Test Accuracy: 80.02 %
Final Precision: 82.12 %
Final Recall: 79.29 %
Final F1-Score: 79.84 %
Model Parameters: 160202
====================================================
```

```
=============== DATASET INFORMATION ===============
Number of Classes : 10
Training Images   : 2964
Validation Images : 740
Testing Images    : 926
```

Sample Traffic Sign Images